FIELD REFERENCE

The Drone Pilot's Cheat-Sheet Pack

Grab-and-Go Cards for Pre-Flight, Settings & Safety

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$17 value

What's Inside

Grab-and-Go Cards for Pre-Flight, Settings & Safety. Written for new pilots who want a printable field reference.

START HERE

How to Use This Pack

These are print-and-keep field cards, not a manual you read once and forget. Each card is dense, self-contained, and built to live in your drone bag — so when you're standing in a field with props in your hand and a gusty sky overhead, you pull the right card and run the check instead of trusting your memory.

Every card that follows stands on its own. There's a card for pre-flight and one for post-flight, a card for camera settings and ND filters, a legal quick-reference, a shot list of cinematic moves, a wind go/no-go table, a battery and return-to-home card, an emergency card, and an export-and-pack-up card. Nothing depends on anything else — grab the one the moment demands.

> HOW TO GET THE MOST FROM THIS PACK
>
> **Print each card, laminate the pre-flight and emergency cards first, and run the pre-flight card every single flight — the routine is what keeps a $1,500 aircraft out of a tree.**

Three ways to use these cards

🖨️ Print & laminate

Each **card starts on its own page**, sized to trim down and slip into your case lid. Laminate them so they survive dew, mud, and the bottom of the bag.

📱 Keep on your phone

Save the PDF to your controller phone or tablet. When you're on-site and unsure, the go/no-go and emergency cards are two taps away.

🔁 Run it every time

The pilots who never lose a drone are the ones who run the same checklist on flight one and flight one thousand. Repetition, not talent, keeps you flying.

> ### 📷 Works with any modern drone
>
> These cards apply to DJI, Autel, and most consumer camera drones. Where a feature has different names by brand, both are given (for example, "Return-to-Home (RTH / Go Home)"). Treat every number as a *starting point* — your aircraft's manual and your local rules always win.

⚠️ Rules change — these cards don't

Airspace, registration thresholds, and local ordinances change and vary by country. The legal card here covers **US/FAA fundamentals only**. Verify the current rules and your specific airspace *before every flight*. A printed card is a memory aid, not legal authority.

CARD 1

Pre-Flight Checklist

Run this top to bottom before you ever spin the props. Most flyaways and crashes trace back to a skipped line on a card exactly like this one. Two minutes here saves you a very bad afternoon.

At home / before you leave

- [x] Aircraft **firmware** and controller firmware updated and matching
- [x] **Batteries charged** — aircraft and controller — and phone/tablet at 80%+
- [x] **SD card** inserted, formatted, and with free space (format in-camera, not on a computer)
- [x] Props inspected for chips, cracks, or stress lines — spares packed
- [x] Weather and wind checked for your window; airspace checked (see Card 4)
- [x] ND filters, spare props, and a landing pad or mat in the bag

On-site, before takeoff

1. **Inspect the airframe.**
   Props seated and tight, arms locked, gimbal clamp removed, lens wiped clean, body free of cracks. Give each prop a gentle spin.

2. **Seat the battery firmly.**
   Listen for the click. A partially seated battery is a top cause of mid-air power loss.

3. **Power up in order.**
   Controller first, then aircraft. Let it boot fully and connect before you touch the sticks.

4. **Wait for a strong GPS lock.**
   Do not launch until the app reports enough satellites and a "Home Point Updated" / "Home Point Recorded" confirmation.

5

Confirm the home point is where YOU are.

Check it on the map. A wrong home point sends the drone "home" to the wrong place on RTH.

6

Set RTH altitude above the tallest obstacle.

See Card 7 for the rule. Do this before takeoff, every time.

7

Check compass & IMU status.

No calibration warnings. Calibrate only if the app asks, and away from metal, cars, and rebar.

8

Clear the area.

No people within ~25 ft of takeoff, no overhead lines, a clean vertical escape path.

The 60-second launch flow

EVERY FLIGHT

1. Airframe & props · **2. Battery seated** · **3. Controller → aircraft on** · **4. GPS lock & home point** · **5. RTH altitude set** · **6. Area clear**

Say each step out loud. Spoken checklists catch the item your eyes skim past.

⚠️ Never skip the home point

If you take off before "Home Point Updated" appears, an automatic Return-to-Home may fly the aircraft to an old or default location — sometimes the last place it locked GPS. Confirm the home point on the map *every* flight.

💡 Hover test

After takeoff, hover at eye level for 5–10 seconds. Listen for odd prop noise, watch for drift, and nudge each stick. If anything feels off, land and diagnose on the ground — not at 300 ft.

CARD 2

Post-Flight Checklist

The flight isn't over when you land — it's over when the gear is safe and logged. Skipping post-flight is how batteries die early and small damage becomes a mid-air failure next week.

Right after landing

1. **Power down in order.**
   Aircraft first, then controller. Let the gimbal settle before you cut power.

2. **Let batteries cool before charging.**
   A battery that's warm from flight should rest **20–30 minutes** and return to near room temperature before it goes on the charger.

3. **Inspect the aircraft.**
   Props for new chips, gimbal for free movement, body for cracks, motors for grit or a gritty feel when spun.

4. **Wipe it down.**
   Lens, sensors, and body. Dust on obstacle sensors causes false warnings and bad avoidance next time.

5. **Log the flight.**
   Date, location, battery cycles used, anything that felt off. Your log is how you spot a failing battery before it fails.

Storage & battery care

- ✓ Reinstall the gimbal clamp before packing
- ✓ Store batteries at **storage charge (~50–60%)** if not flying within ~2 days — most smart batteries self-discharge to this over a set number of days
- ✓ Never store batteries fully charged or fully drained long-term — both shorten their life
- ✓ Store in a cool, dry place, out of direct sun and away from a hot car

- [x] Back up footage off the SD card, then re-format in-camera before the next flight

🛠️ Battery lifespan habit

Lithium batteries hate two extremes: sitting full and sitting empty. Fly them down to ~20–30%, let them cool, and if they'll sit more than a couple of days, leave them near storage charge. This one habit adds dozens of healthy cycles.

📷 Cold weather note

Cold batteries lose voltage fast and can trigger sudden low-power warnings. Warm them to room temperature before flying, keep spares in an inside pocket, and expect shorter flight times below ~40°F / 5°C.

CARD 3

Camera Settings & ND Quick-Set

The look of cinematic drone footage comes from one rule: keep the shutter near **double the frame rate** (the 180° shutter rule) for natural motion blur. In daylight that means an ND filter, because you can't slow the shutter enough on your own. This card sets both.

ND filter by light condition

Light condition	ND strength	Roughly cuts	Target shutter (24–30 fps)
Overcast / heavy cloud	ND4	2 stops	1/50s – 1/60s
Bright cloud / golden hour	ND8	3 stops	1/50s – 1/60s
Hazy sun / partial cloud	ND16	4 stops	1/50s – 1/60s
Bright direct sun	ND32	5 stops	1/50s – 1/60s
Harsh sun / snow / water glare	ND64	6 stops	1/50s – 1/60s

📷 The 180° shutter rule

Set shutter to about **1 / (2 × fps)**. At 24 fps aim for 1/48–1/50s; at 30 fps aim for 1/60s; at 60 fps aim for 1/120s. The ND filter is simply how you hold that shutter without blowing out the sky.

Frame rate & shutter pairings

Frame rate	Use for	Target shutter
24 fps	Cinematic / film look	1/48 – 1/50s
30 fps	Standard, social, real-time	1/60s
60 fps	Smooth motion, 2× slow-mo	1/120s
120 fps	Dramatic slow motion	1/240s

Cinematic daylight baseline

STARTING POINT

Resolution 4K | **FPS** 24–30 | **Shutter** 1/50–1/60s | **ISO** 100 | **ND** match light | **Profile** D-Log / flat

Shoot RAW/DNG photos and a flat/log video profile for the most editing latitude. Keep ISO at base and let the ND control exposure.

💡 Expose for the sky

Bright skies blow out fast and can't be recovered. Set exposure so the brightest clouds keep detail, then lift the shadows in editing. A flat/log profile makes that recovery far easier.

🎓 Fixed shutter, swap the ND

Lock your shutter at the 180° value and never touch it. When the light changes, change the ND filter instead of the shutter. That keeps motion blur consistent across every clip in a sequence.

CARD 4

Legal Quick-Reference (US / FAA)

This card covers US/FAA fundamentals for camera drones. It is a memory aid, not legal advice — rules change and airspace varies by exact location. Read the reminder box, then verify before you fly.

> ⚠️ **Verify locally before every flight**
>
> Airspace class, temporary flight restrictions (TFRs), local ordinances, and registration thresholds change. Check current FAA rules and your **exact location's airspace** in an official app (like B4UFLY or a LAANC provider) *before every flight*. This card does not replace that check.

The fundamentals

Topic	US / FAA fundamental
Registration	Register with the FAA if your drone weighs **0.55 lb (250 g) or more**. Sub-250 g may be exempt for recreational flight — verify.
Recreational vs. Part 107	Flying for fun = recreational rules + **TRUST** test. Any commercial/paid work = **Part 107** certificate required.
Maximum altitude	Stay at or below **400 ft** above ground level in uncontrolled (Class G) airspace.
Visual line of sight	Keep the aircraft in **VLOS** — your own eyes, or a visual observer beside you — at all times.
Controlled airspace	Near airports (Class B/C/D/E) you need authorization, typically via **LAANC**, before you fly.
People & vehicles	Don't fly over people or moving vehicles unless your operation and aircraft specifically permit it.
Remote ID	Most drones that require registration must also **broadcast Remote ID**. Confirm yours complies.
Night & special ops	Night flight, flight over people, and beyond-VLOS have specific requirements — know them before attempting.

> 📷 **Marking your drone**
>
> Registered drones must display the registration number on the exterior. Keep proof of registration (and your Part 107 or TRUST completion) with you when you fly.

💡 When in doubt, don't

No signal on whether an area is legal is itself an answer: don't fly. National parks, many state parks, stadiums, and TFRs (around events, wildfires, VIP movement) are common no-fly surprises. Check first.

CARD 5

Cinematic Moves Shot List

Great aerial footage is a handful of deliberate moves, flown slowly and smoothly. Fly each one *gently* — the smooth, boring-feeling speed is almost always the cinematic one. Work down this list on a shoot and you'll come home with a full sequence.

📤 The Reveal

Rise from behind an object — a ridge, tree line, or building — to unveil the scene beyond it. Slow ascent, steady.

🚀 Fly-Through / Push-In

Move forward past a foreground element toward your subject. Foreground gives the shot depth and scale.

🔄 Orbit / Point of Interest

Circle a subject while keeping it centered. Use the auto POI/Orbit mode for a perfectly smooth arc.

⬇️ Top-Down / Bird's Eye

Gimbal at 90° straight down. Turns roads, coastline, and crowds into graphic patterns.

↩️ Fly-Away Reveal

Start close on a subject, then fly backward and up to reveal how small they are in the landscape.

🎯 Tracking / Follow

Follow a moving subject — car, boat, runner — keeping a steady distance. Use ActiveTrack / Follow mode.

🌀 Vortex / Spiral

Orbit while rising and pushing in. Combines rotation and altitude for a dramatic corkscrew.

🎬 Dolly Zoom-ish Pull

Fly backward while zooming/pushing to compress or stretch depth. Subtle, but striking.

Shoot-the-scene checklist

- ✓ One Reveal to open the sequence
- ✓ One Orbit around the hero subject
- ✓ One Top-Down for a graphic beat
- ✓ One Fly-Away to establish scale
- ✓ One Tracking shot if anything is moving

- [x] Two or three seconds of clean hover before and after each move (editing handles)

> 🎓 **Slow is cinematic**
>
> Turn on your controller's smooth/cine mode to soften stick response, and fly every move slower than feels natural. Jerky, fast footage reads as amateur; slow and deliberate reads as film. When in doubt, halve your speed.

> 💡 **Let each clip breathe**
>
> Hold still for a beat before you start a move and after you finish. Those still handles at each end give your editor clean cut points and make every transition land.

CARD 6

Wind & Weather Go/No-Go

Wind is the quiet killer — it drains batteries, pushes the aircraft off course, and can win a fight against the motors on the way home. Check the wind *at altitude*, which is usually stronger than at ground level, and use this table before every flight.

Condition	Fly	Caution	No-Fly
Wind speed	Under ~15 mph	~15–24 mph	Over ~24 mph (or above your drone's rated max)
Gusts	Light, steady	Gusty, variable	Strong sudden gusts
Rain / snow	None	Light mist (most drones NOT waterproof)	Any real rain or snow
Visibility / fog	Clear, full VLOS	Hazy but VLOS holds	Fog, cloud, or lost VLOS
Temperature	~40–104°F	Near-freezing (short flights)	Below freezing extremes / extreme heat
Lightning	None nearby	Distant, moving away	Any storm within ~10 miles
Sun position	Sun behind/beside you	Low sun in frame	Blinding glare, can't see aircraft

⚠️ Know your aircraft's wind rating

Small drones (sub-250 g) get shoved around at wind speeds a heavier drone shrugs off. Check your model's **maximum wind resistance** spec and treat the table above as a general guide, not a guarantee. When gusts approach that number, land.

💡 Fly out into the wind first

Fly your outbound leg *into* the wind so the trip home is wind-assisted. Fighting a headwind on a low battery is how drones don't make it back. If you feel the aircraft struggling to hold position, come home now.

CARD 7

Battery & Return-to-Home

Two habits prevent most dead-battery losses: the 30% rule and a correct RTH altitude. Burn both into muscle memory and you dramatically shrink your odds of ever losing an aircraft.

THE 30% HABIT

Start heading home at 30% battery — not 10%, not when the app screams. Wind, distance, and cold all eat reserve faster than the percentage suggests. 30% is your "turn back now" line.

Battery reserve, by situation

Situation	Start return at
Close, calm, warm	25–30%
Windy or flying far out	35–40%
Cold weather	40%+ (voltage drops fast)
Land before	Never fly below ~15%

Return-to-Home altitude rule

RTH climbs to a preset altitude, then flies in a straight line home. If that altitude is *below* something between the drone and you — a tree, a building, a hill — RTH flies straight into it. Set the RTH altitude **above the tallest obstacle** in your entire flight area, before takeoff.

- [x] RTH altitude set higher than the tallest tree, tower, or building around you
- [x] Home point confirmed at your actual location (see Card 1)
- [x] You know how to **cancel RTH** and fly manually if its path looks wrong
- [x] Obstacle avoidance on — but never trusted blindly (it can miss thin branches and wires)

⚠️ RTH is not a magic button

Return-to-Home flies a straight line at a fixed height. It does not route around obstacles below its set altitude, and avoidance sensors miss wires, netting, and thin branches. If you can still see and control the aircraft, flying it home manually is often safer.

🎓 Know your real range on this battery

Advertised range assumes calm air and a fresh battery. Your usable there-and-back distance is far shorter — especially into wind. Log how far out you were when the 30% warning hit, and let that real number set your limits next time.

CARD 8

Emergency Card

When something goes wrong, panic costs aircraft. Find the situation, run the numbered steps in order, calmly. Keep this card laminated and near the top of your bag.

Flyaway (drone drifts off, not responding)

1. **Don't yank the sticks.**
   Center them and give it a second — sudden inputs during a GPS glitch make it worse.
2. **Trigger Return-to-Home.**
   Hit the RTH button on the controller or in the app.
3. **If RTH fails, switch to ATTI / manual.**
   Fly it back by hand using your visual on the aircraft, nose pointed back toward you.
4. **Reduce altitude to re-establish control.**
   Lower, closer air often restores a clean signal and orientation.
5. **If it's truly lost, note last position & heading.**
   Screenshot the map and the last coordinates for recovery.

Signal loss (video/control drops out)

1. **Stop and hold position.**
   Most drones auto-hover or auto-RTH on signal loss — let the failsafe work.
2. **Raise the controller and antennas.**
   Point antennas *broadside* (not tips) at the aircraft; get clear of buildings and your own body.
3. **Walk toward its last known position.**
   Closing distance and clearing obstructions usually restores the link.

4 **Wait for RTH to bring it home.**

If your failsafe is set to RTH, it should climb and return — keep eyes up to take over.

Low battery (warning fires mid-flight)

1 **Turn toward home immediately.**

Don't finish the shot. The clock is real.

2 **Descend as you approach.**

Lower altitude uses less power for the last leg and shortens the trip down.

3 **Take the nearest safe landing spot.**

A safe landing in the wrong field beats a dead drone reaching for the right one.

4 **Let auto-land finish.**

If the critical warning triggers auto-landing, don't fight it — guide it laterally to clear ground and let it set down.

> ⚠️ **Protect people first, aircraft second**
>
> If the choice is a hard landing or flying over a crowd, take the hard landing. A replaced drone is cheap; an injured person is not. Steer away from people, roads, and water in every emergency.

CARD 9

Export & Pack-Up

The last card closes the loop: get the footage out at the right settings, then pack the gear so it's ready for next time. These export recipes are starting points — match resolution and frame rate to how you shot.

Export recipes

YouTube / landscape

16:9

Res 3840×2160 (4K) · **FPS match source** · **Codec H.264 / H.265** · **Bitrate ~45–65 Mbps (4K)**

Export at your capture frame rate. Higher bitrate holds up better against YouTube's re-compression.

Vertical / Reels & Shorts

9:16

Res 1080×1920 · **FPS 30** · **Codec H.264** · **Length < 60s**

Reframe from your 4K source so you can crop to vertical without losing resolution. Keep the subject in the center-safe zone.

Client delivery

MASTER

Res 4K · **FPS match source** · **Codec H.265 / ProRes** · **Bitrate high / near-lossless**

Deliver a high-bitrate master plus a compressed review copy. Include a color-graded version and, if asked, the flat/log originals.

> 💡 **Shoot 4K even for 1080p delivery**
>
> Capturing in 4K and delivering 1080p lets you crop, reframe, and stabilize in the edit with room to spare — and the downscaled result looks sharper than native 1080p.

Final pack-up checklist

- [x] Footage backed up off the SD card (two copies before you delete anything)

- [x] Gimbal clamp on, props inspected one last time
- [x] Batteries at storage charge if not flying soon; cooled before packing
- [x] Controller sticks stowed, cables coiled, spare props and NDs accounted for
- [x] Flight logged; anything odd noted for next time
- [x] Bag zipped, nothing left in the field (props and prop guards go missing fast)

> 🛠️ **Two cards to keep closest**
>
> Of the whole pack, laminate the **pre-flight** and **emergency** cards first and keep them at the top of your bag. One prevents the disaster; the other saves the day when it happens anyway.

REMEMBER

The pilots who never lose a drone aren't the most talented — they're the ones who ran the pre-flight card before every single flight, so make sure you're ready before you fly.

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